

Voice Recognition based Operation Theatre Assistive Robot

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Abstract

This paper describes the design and analysis of an assistive robot with a 4-axes articulated robotic manipulator. The current work is undertaken by considering various medically used robots to design and fabricate a 4-DOF robotic arm. A forward kinematic model has been presented in order to determine the end effector's position and orientation. This analysis is useful for path tracking and trajectory following of a surgical robot or assistive robot with the 'pick and place' application of surgical tools using voice recognition. Based on this analysis, a researcher can develop a path-tracking behavior of an end-effector in a complicated workspace. We have succeeded in recognizing the voice of a person (in a noise-free environment) and hence picking the same tool as spoken by the person and placing it in his hands using color recognition methods and then returning to its initial position.

Keywords: 4-DOF robotic arm; forward kinematic model; trajectory planning; surgical tools; voice recognition.

1. Introduction

In the age of cutting-edge technologies, applied autonomy has turned out to be well-known, and helpful and has accomplished extraordinary triumphs in a few fields of mankind. Robots are very powerful elements of today's industry. They can perform different tasks and operations, are accurate, and do not require common safety and comfort elements that humans need. Nonetheless, it requires much exertion and numerous assets to make a robot work legitimately. Robotics is a relatively new field of modern technology that crosses traditional engineering boundaries.

Robots, which could be ruinous or non-damaging, perform operations that would have been exceptionally dull for people to perform. They are fit for performing monotonous undertakings all the more rapidly, inexpensively, and precisely than people. The modern robots of today may not look even a little bit like a person albeit all the exploration is coordinated to give increasingly human and human-like highlights and super-human capacities in these.

The first industrial robot was Unimates developed by George Devol and Joe Engel Berger in the late 50s and early 60s. The first patents were by Devol but

Berger formed Unimation which was the first market robot. Portable mechanical autonomy moved into its very own in 1983 when Odetics presented a six-legged vehicle that was equipped for moving over items. This robot could lift over 5.6 occasions its own weight stopped and on 2.3 occasions its weight moved. Another exploration field in applying autonomy enlivened by organic frameworks has emerged. The technological developments have made it possible for engineers and robot designers to look for solutions in nature and look forward to achieving one of their most attractive goals to develop a humanoid robot. Bio robotics is historically connected to service robotics. These robots are conceptualized in an unexpected way in comparison to mechanical robots. Their task is usually to help humans in diverse activities from house cleaning to carrying out a surgery or playing the piano to assisting the disabled and the elderly. It is normal that humanoid robots will have the capacity to speak with people and different robots; encourage robot programming, increment their adaptability and flexibility for executing diverse errands; gain as a matter of fact; and adjust to various assignments and situations or change of place [1].

In the usage of natural conduct frameworks, the replication of human attributes is perhaps the appropriate response in each set of advancements in applying autonomy. The research in anthropomorphic robotics has advanced to the development of anthropomorphic components for humanoid robots like anthropomorphic visual and tactile sensors, anthropomorphic actuators, and anthropomorphic computing techniques. Replicating the functionality of the human brain is one of the hardest challenges in Bio-robotics and in general still one of the most difficult objectives.

2. Literature Review

When analyzing the information sources, high activity was revealed among the studies of simple manipulator control systems, which are based on actuators with controlled rigidity to ensure human safety. The literature review is based according to the different parameters of a robotic arm.

The most important parameters based on which the robotic manipulator has been designed are axes, degrees of freedom, working envelop kinematics, payload, speed and acceleration, accuracy and repeatability, motion control, and drive.

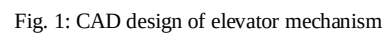
Manipulators are grouped into different modules depending on the type of joints and their combinations. There are five different major classes of manipulators. Cartesian manipulator consists of all three prismatic joints. A cylindrical manipulator consists of two prismatic and a single revolute joint. Spherical and SCARA types consist of two revolute and one prismatic joint with a difference in symmetry. Lastly, an articulated manipulator consists of all three revolute joints. Manipulators other than these are the combination of these five classes with different kinematics and path trajectories.

In 2002, the students of the University of Pittsburgh, the University of Michigan, and Carnegie Mellon University developed a mobile robot assistant for elderly people to go through their daily life routines, especially with chronic diseases. The two major tasks to be fulfilled by PEARL were: (i) reminding elderly patients about daily routines like eating food and taking medicines, etc. (ii) guiding them through their daily environment. A prototype of PEARL in working mode is implemented in the Longwood Retirement community in Oakmont, PA [3].

3. Design and Calculations

Surgical tools are to be delivered in the hands of surgeons through the detection of the hand. Tools will be placed in the trays mounted on an elevator mechanism. As the name of the tool suggests, the elevator positions itself in such a way that the robotic arm can grasp the tool. Currently, we use four tools to call, pick, and place. CAD design with proper dimensions is shown below:

Diameter of rod = 1 cm



Link i	Notation Used	Length (ft.)
Link 1	L_1	0.5
Link 2	L_2	0.75
Link 3	L_3	0.7
Link 4	L_4	0.55

The manipulator used in this project is a 4-axe, 4-DOF robotic arm, which is the focus of this project. Articulated configuration is employed as we need revolute joints and hence revolute motion. Link lengths are estimated according to the span of the planar workspace as listed in Table I, which is described in the further section. CAD design is shown in Fig. 2.

At the 95th percentile, female and male average heights are 5.00 ft. and 5.57 ft. respectively. Integrating the average population data with working height guidelines from laparoscopic studies implies that the operating table height should range from 2.08 ft. to 2.5 ft. above the floor.



TABLE II. Average height and torso

The average height of male	5.57 ft.
Torso of the average male	1.54 ft.

Dimensions of our planar workspace are 1.5 X 1.5 ft.

The height of the operating table from the floor is 2.08 ft. to 2.5 ft [4].

3.4 Design concept

The key points in the R&D of this robot are as follows:

- Design and fabrication of a robotic arm with four degrees.
- Hand detection by color recognition.
- Restricted Voice recognition.

The major task in this project was to design a highly accurate robotic manipulator arm, which was difficult to achieve. Including the end-effector, there are 5 servo motors. Each motor was programmed to move in a definite direction with definite angles to achieve the required positioning. A series of testing and troubleshooting took place and different types of hurdles and barriers were encountered during the process, like motor damage, friction factors, and wiring issues.

Secondly, the coupling of the voice recognition module with the elevator movement and the manipulator movement was a vital task to accomplish. All these tasks were achieved through coding using Arduino Uno and Atmega.

The simple plan of this project was to make a prototype to work in an operation theatre environment, which is a noise-free environment. This prototype can recognize the voice of the surgeon and understand the name of the tool spoken by the surgeon through a series of training of VRM. As soon as it understands the tool name the respective tool tray comes to the top by using the elevator mechanism. Meanwhile, the arm moves in the specific manner as programmed and picks up the tool and moves towards the doctor's hand, and through color recognition of the gloves, it places precisely in the hands of the doctor.

4. Forward Kinematics

As for the dimensions of the links, the lengths (L_1 , L_2 , L_3 , and L_4) of the links were established. Robo Analyzer software was used for the accurate calculation of forward kinematics. Table III shows the Denavit-Hartenberg parameters obtained from the mechanical design and configuration of the frames.

TABLE III. Denavit-Hartenberg parameters

Link i	a_i	α_i	d_i	Θ_i
1	0	-90	L_1	Θ_1^*
2	L_2	0	0	Θ_2^*
3	L_3	0	0	Θ_3^*
4	L_4	0	0	Θ_4^*

Using Denavit-Hartenberg (DH) convention, the coordinate frames for the manipulator are assigned as shown in Fig. 3.

Direct kinematics of a robot system is described by a transformation matrix T , which relates both the position and orientation of the endpoint (tool) of the robot with respect to a fixed reference (frame) coordinate system, which generally is the physical basis or frame [5]. This relationship is made using the particular homogeneous transformations between joints, that involve translations and rotations. However, it is possible to obtain complete homogeneous transformations that involve all the joints in a general form [6]. In our case:

$$T_0^4 = T_1^0 \cdot T_1^2 \cdot T_2^3 \cdot T_3^4 \quad (1)$$

The D-H parameters are assigned to the respective frames for each pair of joints, resulting in the following matrix.

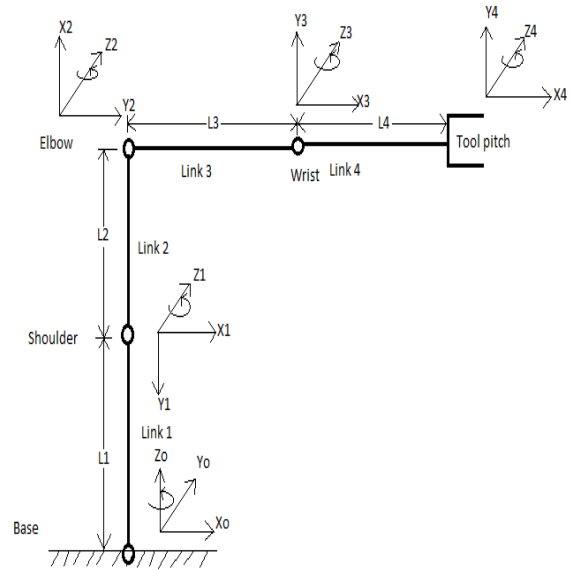


Fig. 3: Free body diagram

$$T_{10} = \begin{bmatrix} C_1 & 0 & -S_1 & 0 \\ S_1 & 0 & C_1 & 0 \\ 0 & -1 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{12} = \begin{bmatrix} C_2 & -S_2 & 0 & C_2 \cdot L_2 \\ S_2 & C_2 & 0 & L_2 \cdot S_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{23} = \begin{bmatrix} C_3 & -S_3 & 0 & C_3 \cdot L_3 \\ S_3 & C_3 & 0 & C_3 \cdot L_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{34} = \begin{bmatrix} C_4 & -S_4 & 0 & C_4 \cdot L_4 \\ S_4 & C_4 & 0 & C_4 \cdot L_4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Total transformation is given as:

$$T_0^4 = \begin{bmatrix} m_x & n_x & o_x & p_x \\ m_y & n_y & o_y & p_y \\ m_z & n_z & o_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$m_x = C_4 \cdot (C_1 \cdot C_2 \cdot C_3 - C_1 \cdot S_2 \cdot S_3) - S_4 \cdot (C_1 \cdot C_2 \cdot S_3 + C_1 \cdot C_3 \cdot S_2) \quad (2)$$

$$n_x = -C_4 \cdot (C_1 \cdot C_2 \cdot S_3 + C_1 \cdot C_3 \cdot S_2) - S_4 \cdot (C_1 \cdot C_2 \cdot C_3 - C_1 \cdot S_2 \cdot S_3) \quad (3)$$

$$o_x = -S_1 \quad (4)$$

$$p_x = C_1 \cdot C_2 \cdot L_2 - C_4 \cdot L_4 \cdot (C_1 \cdot C_2 \cdot S_3 + C_1 \cdot C_3 \cdot S_2) + C_4 \cdot L_4 \cdot (C_1 \cdot C_2 \cdot C_3 - C_1 \cdot S_2 \cdot S_3) - C_1 \cdot C_3 \cdot L_3 \cdot S_2 + C_1 \cdot C_2 \cdot C_3 \cdot L_3 \quad (5)$$

$$m_y = C_4 \cdot (C_2 \cdot C_3 \cdot S_1 - S_1 \cdot S_2 \cdot S_3) - S_4 \cdot (C_2 \cdot S_1 \cdot S_3 + C_3 \cdot S_1 \cdot S_2) \quad (6)$$

$$n_y = -C_4 \cdot (C_2 \cdot S_1 \cdot S_3 + C_3 \cdot S_1 \cdot S_2) - S_4 \cdot (C_2 \cdot C_3 \cdot S_1 - S_1 \cdot S_2 \cdot S_3) \quad (7)$$

$$o_y = C_1 \quad (8)$$

$$p_y = C_2 \cdot L_2 \cdot S_1 - C_4 \cdot L_4 \cdot (C_2 \cdot S_1 \cdot S_3 + C_3 \cdot S_1 \cdot S_2) + C_4 \cdot L_4 \cdot (C_2 \cdot C_3 \cdot S_1 - S_1 \cdot S_2 \cdot S_3) + C_2 \cdot C_3 \cdot L_3 \cdot S_1 - C_3 \cdot L_3 \cdot S_1 \cdot S_2 \quad (9)$$

$$m_z = -C_4 \cdot (C_2 \cdot S_3 + C_3 \cdot S_2) - S_4 \cdot (C_2 \cdot C_3 - S_2 \cdot S_3) \quad (10)$$

$$n_z = S_4 \cdot (C_2 \cdot S_3 + C_3 \cdot S_2) - C_4 \cdot (C_2 \cdot C_3 - S_2 \cdot S_3) \quad (11)$$

$$o_z = 0 \quad (12)$$

$$p_z = L_1 - L_2 \cdot S_2 - C_2 \cdot C_3 \cdot L_3 - C_3 \cdot L_3 \cdot S_2 - C_4 \cdot L_4 \cdot (C_2 \cdot S_3 + C_3 \cdot S_2) - C_4 \cdot L_4 \cdot (C_2 \cdot C_3 - S_2 \cdot S_3) \quad (13)$$

5. Conclusion

We have developed a prototype of robot as shown in Fig. 6, to assist in the operation theatre environment. This robotic hand has succeeded in understanding the tool name and recognizing the voice, and passing the tool to the surgeon in approximately 8 sec. To the best of our knowledge, at this level, we have tried to make it highly accurate and precise with some uncertainties. Due to shortage of time, which is about a year comprising of research and development of this project.

The prospects of this project include patient monitoring systems, minimally invasive and robotic surgery, next generation imaging, bionic robot using AI and rehabilitation robots. It is a vast project to continue with and we are proud members to be a part of this effort.

6. Acknowledgement

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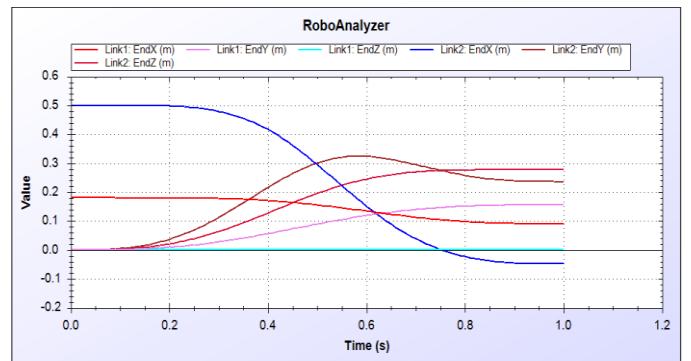


Fig. 4: Graphs for the motion of first two links

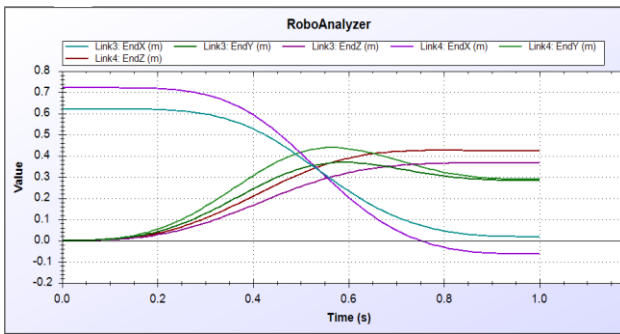


Fig. 5: Graphs for next two links

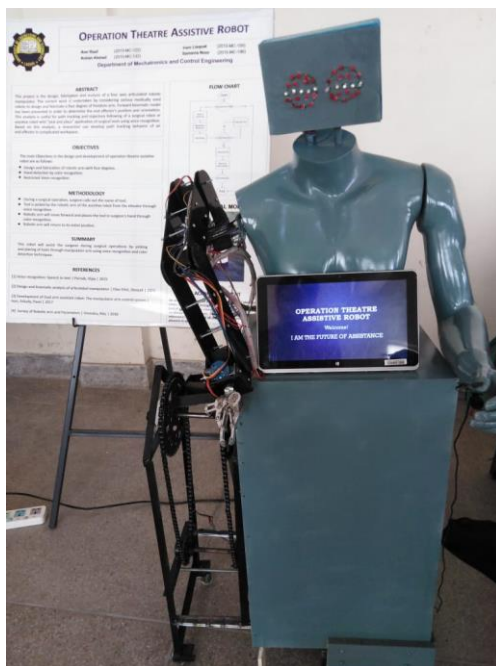


Fig. 6: Final Prototype

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